Right now, we're sitting at a ~15 minute computation time for the Barnes-Hut computation. My goal is to hopefully decrease this time down to ~10 seconds (about an order of magnitude or two decrease).

From the "Many-Body Tree Methods in Physics" book, there are a TON of optimizations I can make. My main goal is to try and implement in as many as possible, such as vectorizing and parallelizing the Barnes-Hut implementation, implementing in the Fast Multipole Method, and then also using CUDA for GPU computing, as described by the two below papers:

Reading List

1. Many-Body Tree Methods in Physics (Chapters 2, 4, and 7) by Gibbon and Pfalzner
2. An Efficient CUDA Implementation of the Tree-Based Barnes Hut n-Body Algorithm[1] by Burtscher and Pingali
3. Introduction to Fast Multipole Methods[2] by Chen
4. Real-time N-body Simulation with Barnes-Hut Algorithm and CUDA [3] by Wu
5. A short course on fast multipole methods [4] by Beatson and Greengard

[1]ScienceDirect
[2]UC Irvine
[3]GitHub
[4]New York University